

Laxatives & Anti-Diarrhoeal Agents

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Learning Objectives

1. Categorize Laxatives

List and classify the primary types of laxative agents based on pharmacological class.

2. Understand MOA

Master the mechanisms of action for all major laxative classes.

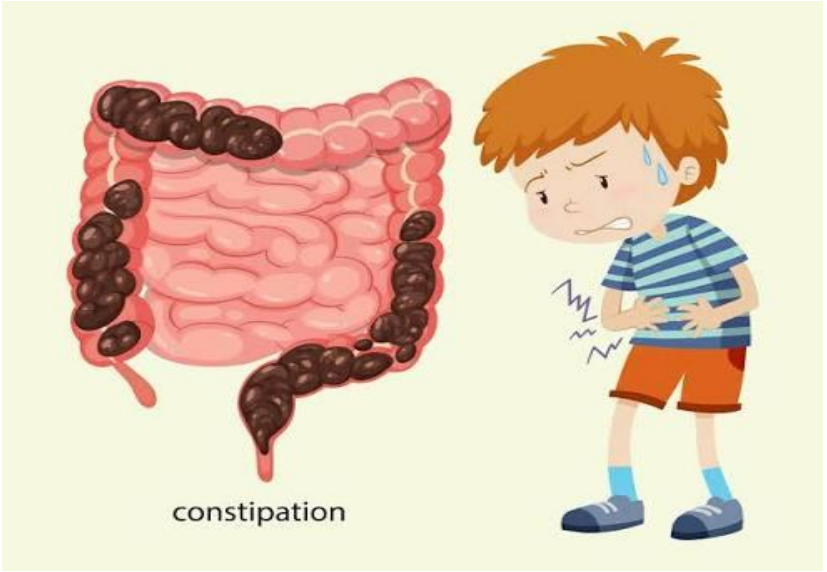
3. Anti-Diarrhoeals

Describe anti-diarrhoeal agents along with their detailed mechanisms of action.

Constipation & Initial Management

Practical Definition

A reduced frequency or ease of stool passage relative to what is regarded as normal or expected for that individual.



First-Line Non-Drug Therapy/ non pharmacological

- **Fiber-Rich Diet:** Fruits, vegetables, whole-meal bread.
- **Hydration:** Increased fluid intake.
- **Dietary Restrictions:** Minimize tea and coffee.
- **Physical Exercise:** Activates abdominal muscles and intestinal peristalsis.

PHARMACOTHERAPY

Laxatives: Core Principles & Risks

Primary Purpose:

Laxatives accelerate the movement of food and waste material through the gastrointestinal tract.

Drug Loss

Accelerated intestinal transit increases loss of pharmacologic effect for poorly absorbed, delayed-acting, or extended-release oral formulations.

Electrolyte Balance

Chronic or excessive laxative administration frequently leads to significant electrolyte imbalances.

Dependency Risk

Prolonged reliance carries a high risk of drug dependency, leading to severe loss of normal intestinal motility.

CLINICAL MANAGEMENT

Indications & Contraindications

Indications

- **Constipation:** Short-term use only (prevents dependence).
- **Toxic Substance Excretion:** Rapid evacuation from the GIT.
- **Diagnostic Preparation:** Prior to X-ray or colonoscopy.
- **Anal Conditions:** Anal fissures or hemorrhoids (piles).
- **Postoperative Recovery:** Post-hemorrhoidectomy to avoid straining.

Contraindications

- **Undiagnosed Abdominal Pain:** High risk of masking acute disease.
- **Inflammatory Bowel Disease (IBD):** May trigger intestinal perforation.
- **Organic Obstruction:** Absolutely contraindicated in mechanical GIT blockages.

Warning: Laxatives in acute surgical abdomen can lead to fatal intestinal rupture.

PHARMACOLOGICAL TAXONOMY

Classification of Laxative Agents

Mechanism Class	Generic Name	Trade Name Examples
Irritants & Stimulants	Bisacodyl, Castor oil, Senna	Correctol, Dulcolax, Ex-Lax, Senokot
Bulk Laxatives	Methylcellulose, Psyllium	Citrucel, Metamucil
Saline & Osmotics	Lactulose, Mg citrate, Mg hydroxide, Polyethylene glycol	Constulose, Milk of Magnesia, GoLYTELY, Miralax
Stool Softeners	Docusate	Colace
Lubricant Laxatives	Glycerin suppositories, Mineral oil	Generic preparations

CLASS A

Irritants & Stimulants

1. Senna

Action: Widely used stimulant. Oral intake causes bowel evacuation within 8 to 10 hours.

Mechanism: Promotes active water and electrolyte secretion into intestinal lumen.

2. Bisacodyl

Forms: Suppositories & enteric-coated tablets.

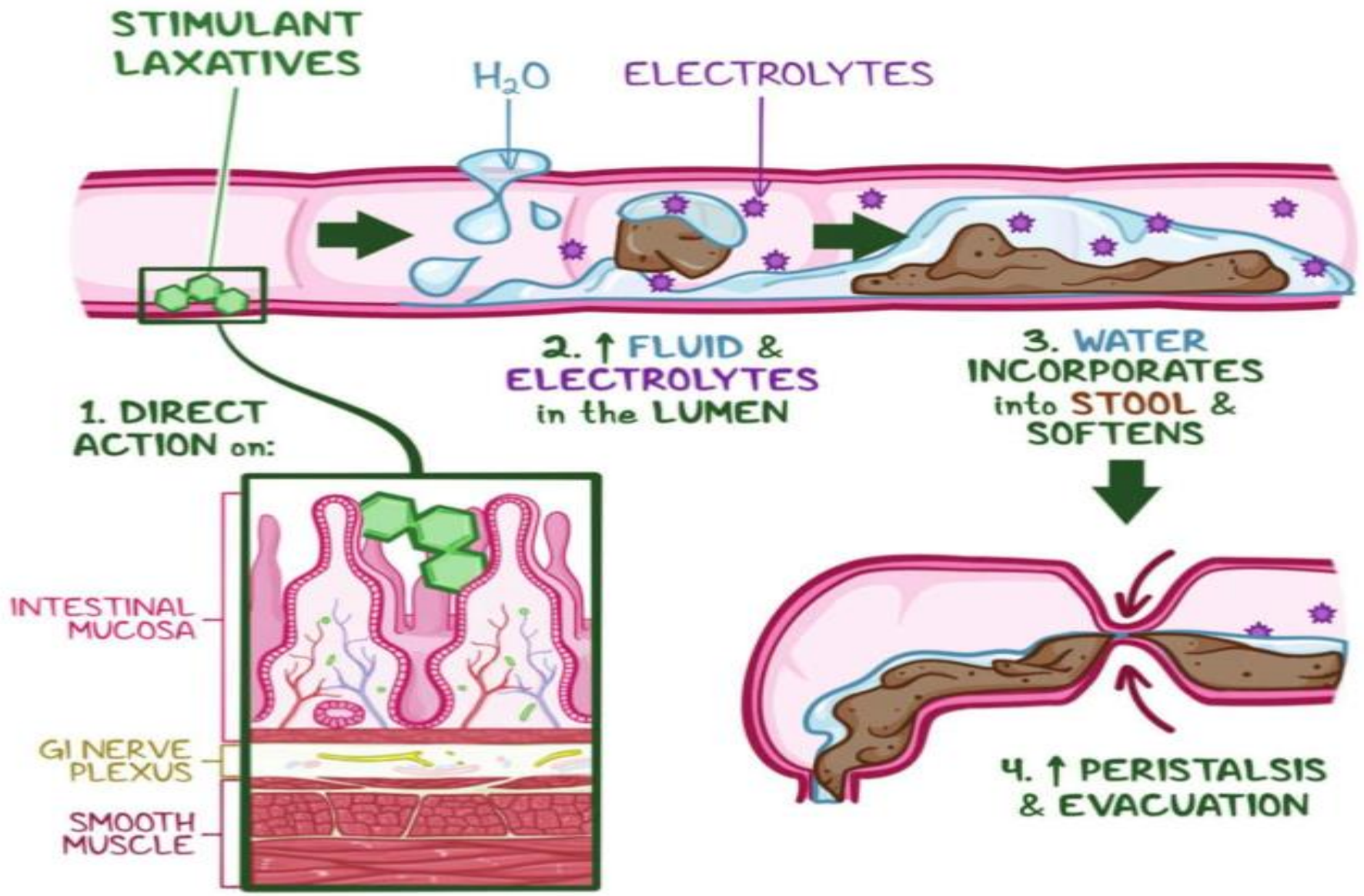
Mechanism: Potent colon stimulant acting directly on nerve fibers in colonic mucosa to induce motility.

3. Castor Oil

Mechanism: Hydrolyzed in small intestine to *ricinoleic acid*, an irritating agent increasing peristalsis.

⚠ **Pregnancy Warning:** Contraindicated as it stimulates uterine contractions.

Stimulant Laxatives



CLASS B

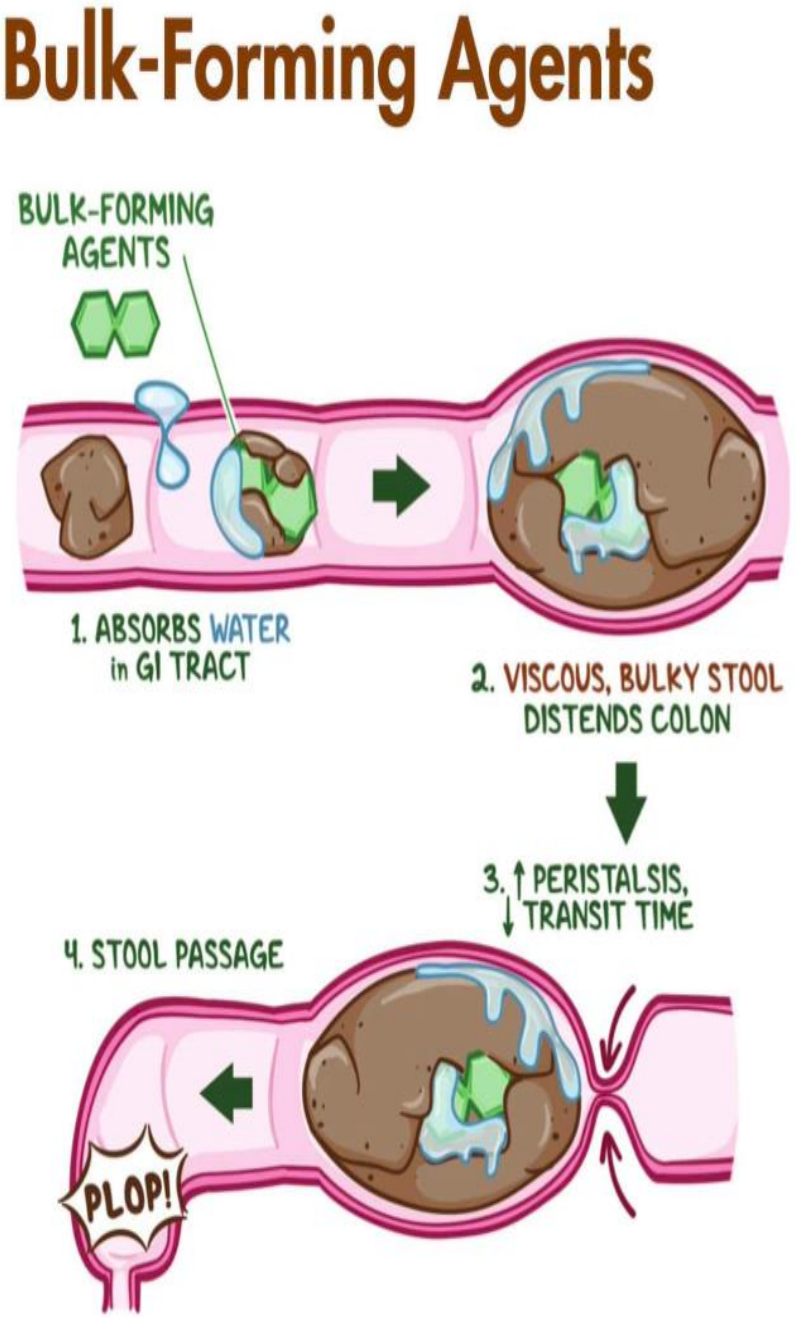
Bulk Laxatives

Mechanism & Hydrophilic Properties

- **Composition: Formed from hydrophilic colloids derived from indigestible plant parts.**
- **Gel Formation: Forms gels in the large intestine, causing water retention and bowel distension.**
- **Peristalsis Boost: Distension reflexively stimulates intestinal peristaltic activity.**
- **Key Agents: Methylcellulose, psyllium seeds, bran.**

Immobile Patient Warning

Use cautiously in bedridden or immobile patients due to high potential for causing mechanical intestinal obstruction.



CLASS C

Saline & Osmotic Laxatives

Saline Cathartics

Agents: **Magnesium citrate & Magnesium hydroxide.**

MOA: Nonabsorbable salts that hold fluid in the bowel via osmosis, leading to distension and defecation within hours.

Polyethylene Glycol

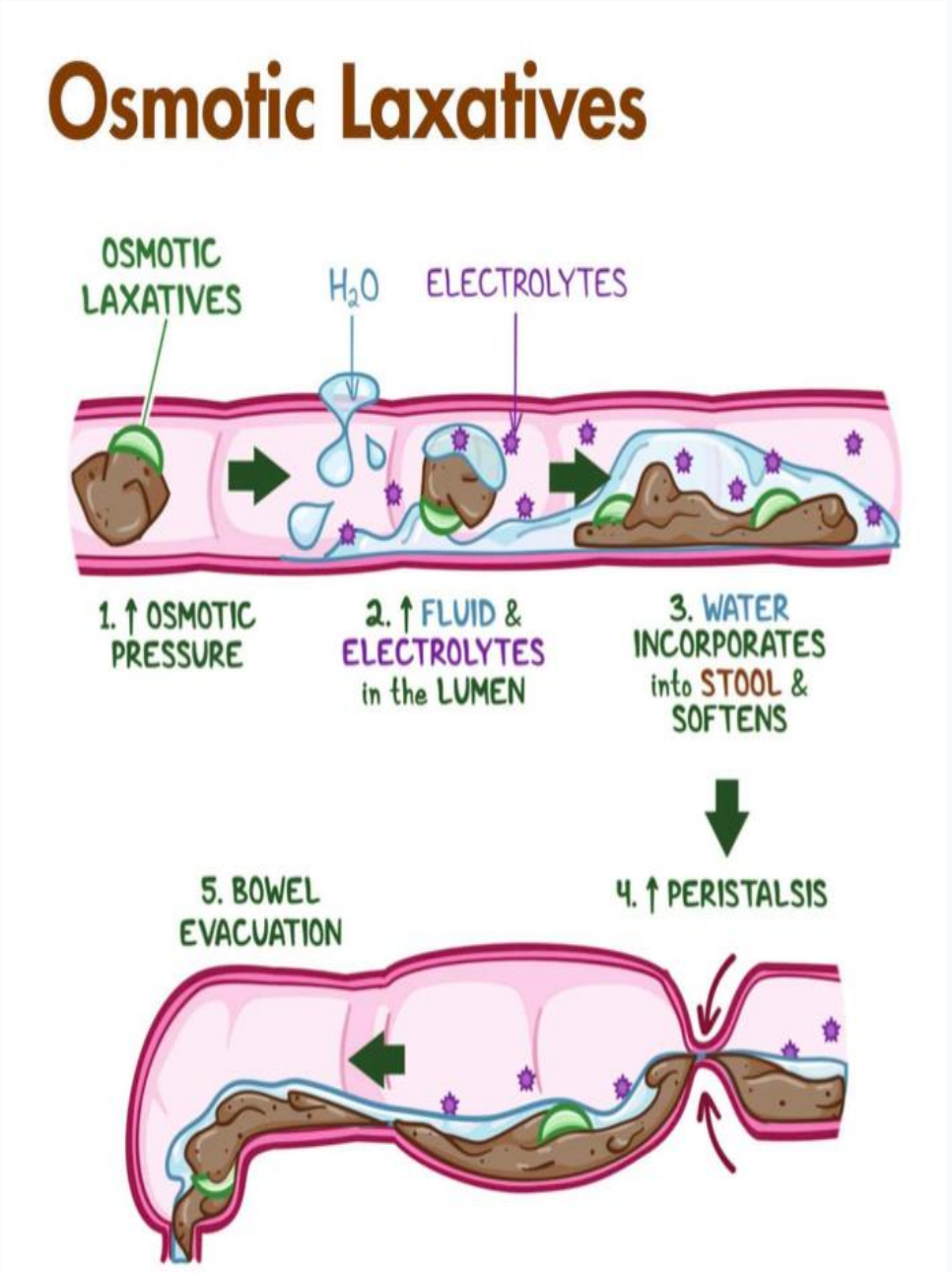
Agents: **polyethylene glycol** PEG solutions

Clinical Application: Electrolyte solutions used as colonic lavage to clear the gut prior to radiological or endoscopic procedures.

Lactulose

Properties: Semisynthetic disaccharide sugar acts as an osmotic laxative.

MOA: Resists GI enzymes; colonic bacteria metabolize it to lactic, formic & acetic acids
This increases osmotic pressure, causing fluid accumulation, colon distension, and soft stools and defecation.



CLASSES D & E

Stool Softeners (Emollient and lubricant)

Stool Softeners (Emollients)

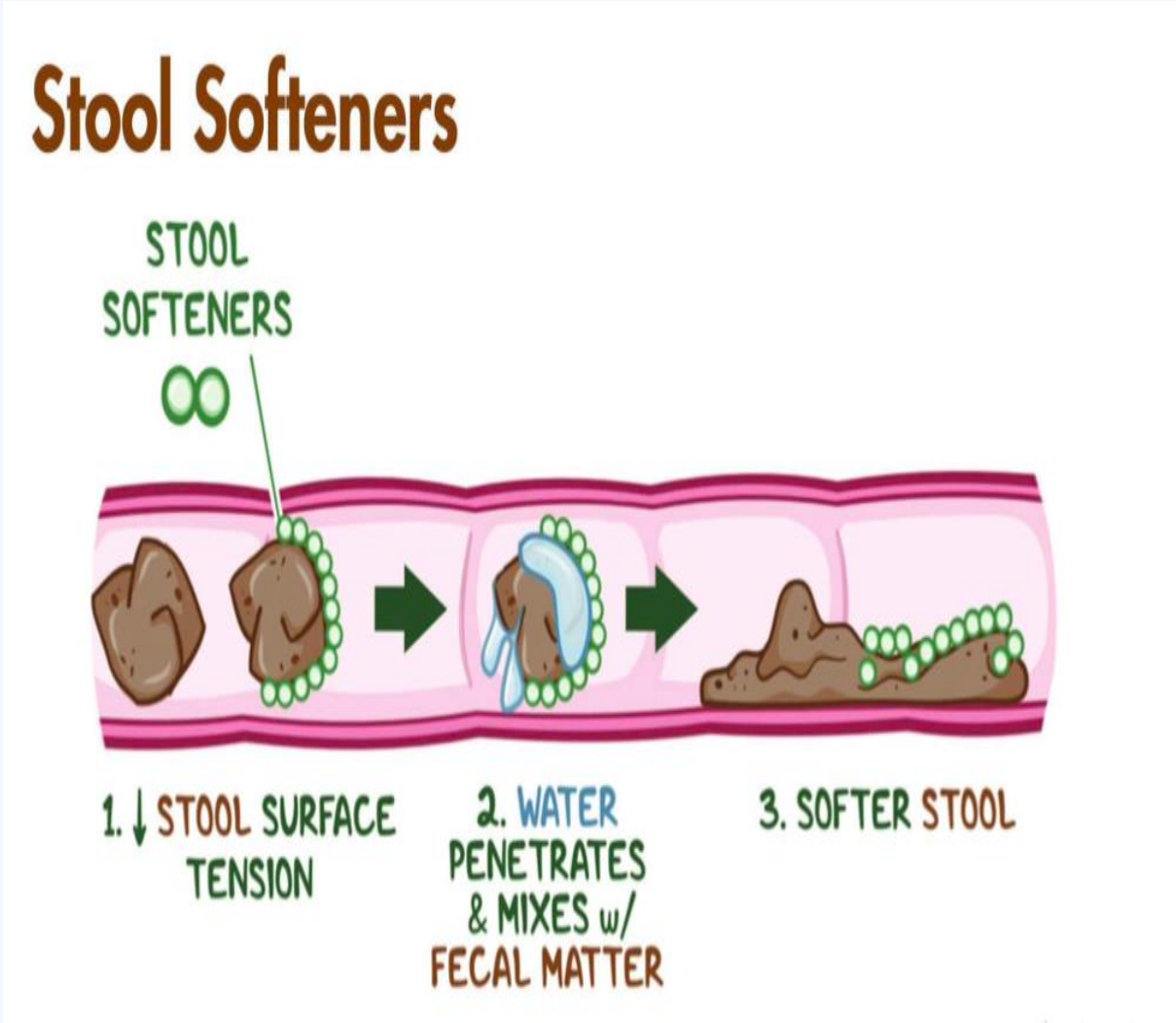
- Agents: Docusate sodium and Docusate calcium.
- Mechanism: Surface-active surfactants that emulsify with stool, producing softer feces for easier passage.
- Clinical Use: Takes days to become effective; used primarily for prophylaxis rather than acute treatment.

Lubricant Laxatives

- Agents: Mineral oil and Glycerin suppositories.
- Mechanism: Lubricates the intestinal wall and facilitates passage of hard stool.

Administration Safety

Oral mineral oil must be taken in an **upright position** to prevent aspiration into lungs and severe **lipid pneumonia**.



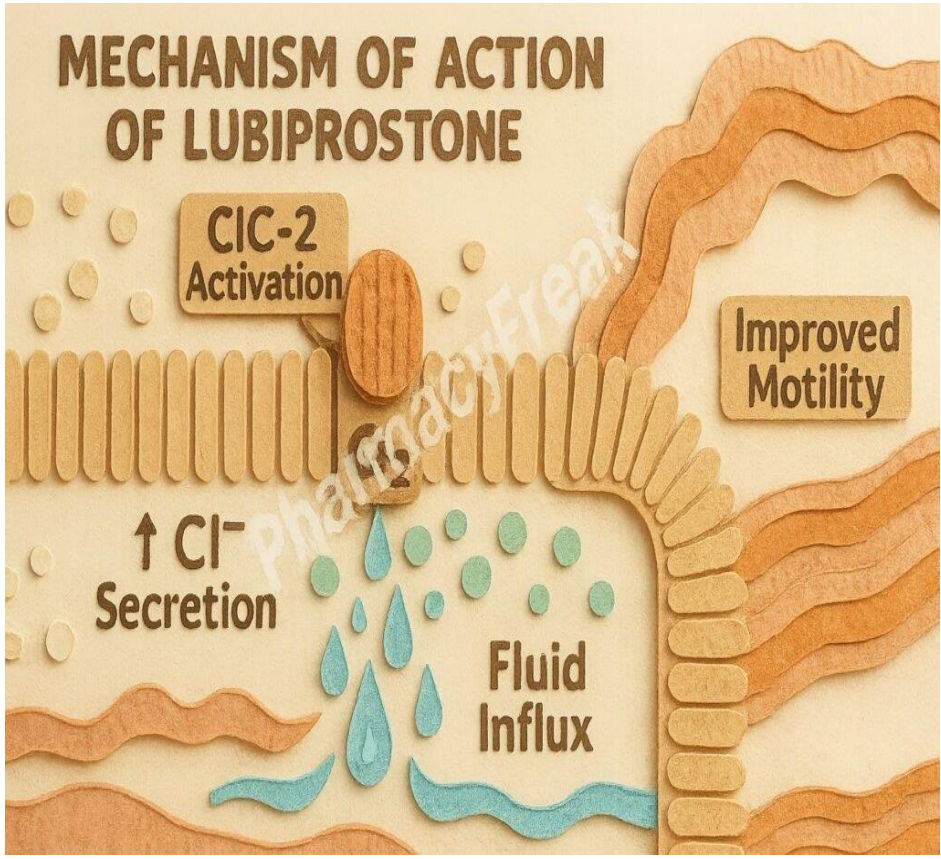
CLASS F

Chloride Channel Activators

Lubiprostone

Currently the sole targeted agent in this pharmacological class.

- **Mechanism of Action: Activates luminal chloride channels, driving fluid secretion into the intestinal lumen to ease stool passage.**
- **Electrolyte Balance: Increases fluid excretion with minimal alteration in overall systemic electrolyte equilibrium.**
- **Chronic Indications: Approved for chronic constipation; carries no risk of tolerance or dependency.**
- **Minimal Drug Interactions: Rapidly metabolized in the stomach and jejunum before systemic distribution.**



ANTI-DIARRHOEAL THERAPY

Pathophysiology & Management

⚠ Pathophysiologic Drivers

1. Increased GI Motility

2. Decreased Fluid Absorption

Are major factors in diarrhoea

3 Core Approaches in Severe Acute Diarrhoea

1. Fluid Balance

Maintenance of fluid & electrolyte equilibrium (ORS).

2. Anti-Infectives

Specific antimicrobial agents for infectious etiology.

3. Spasmolytics

Motility reduction & symptomatic antidiarrhoeal agents.

ANTI-DIARRHOEAL CLASSES

A. Antimotility Agents

Diphenoxylate & Loperamide

- **Opioid Analogs:** Structure-related to meperidine; display selective opioid-like actions on the gut.
- **Presynaptic Activation:** Stimulate presynaptic opioid receptors in the enteric nervous system.
- **ACh Inhibition:** Inhibits acetylcholine release, significantly decreasing smooth muscle peristalsis.
- **CNS Profile:** Lacks systemic analgesic effects at typical clinical doses.

Formulations

- Diphenoxylate + Atropine — LOMOTIL
- Loperamide — IMODIUM A-D

Toxic Megacolon Risk

Contraindicated in young children and patients with severe colitis due to the high risk of triggering toxic megacolon.

B. Adsorbents

Agents: Aluminum hydroxide, Activated charcoal, Methylcellulose.

- **Mechanism: Adsorbs intestinal toxins and microorganisms and/or coats and protects the mucosa.**
- **Efficacy: Significantly less effective than antimotility agents.**
- **Interaction: Can adsorb and interfere with the absorption of other co-administered oral drugs.**

C. Fluid Transport Modifiers

Agent: Bismuth Subsalicylate PEPTO-BISMOL

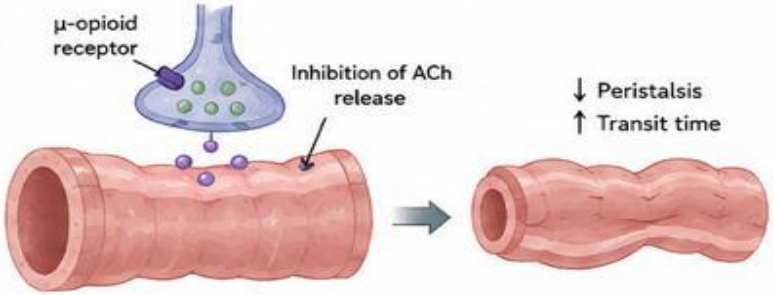
- **Indication: First-line choice for Traveler's Diarrhoea.**
- **MOA: Decreases fluid secretion via salicylate anti-inflammatory component and mucosal coating.**
- **Side Effects: Harmless darkening of the tongue and black stools due to bismuth sulfide formation.**

MECHANISM OF ACTION OF ANTI DIARRHEAL DRUGS

1 ANTI MOTILITY AGENTS
(Diphenoxylate, Loperamide)

MECHANISM
Decrease intestinal motility and transit time, allowing greater absorption of water and electrolytes.

SITE & ACTION
- Act on μ-opioid receptors in the myenteric plexus of the intestinal wall.
- Inhibit release of acetylcholine and other excitatory neurotransmitters.



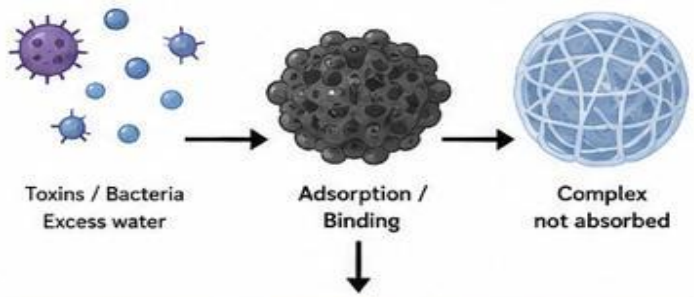
RESULT
↑ Absorption of water and electrolytes
→ Firmer stools, ↓ frequency of stools

EXAMPLES
- **Loperamide** – Peripheral action, does not cross BBB
- **Diphenoxylate** – Peripheral action, often combined with atropine to discourage abuse

2 ADSORBENTS
(Activated Charcoal, Methyl Cellulose)

MECHANISM
Adsorb toxins, bacteria, and excess water within the intestinal lumen.

SITE & ACTION
- Non-systemic agents that bind to toxins, microorganisms and other harmful substances.
- Methyl cellulose also absorbs excess water and increases stool bulk.



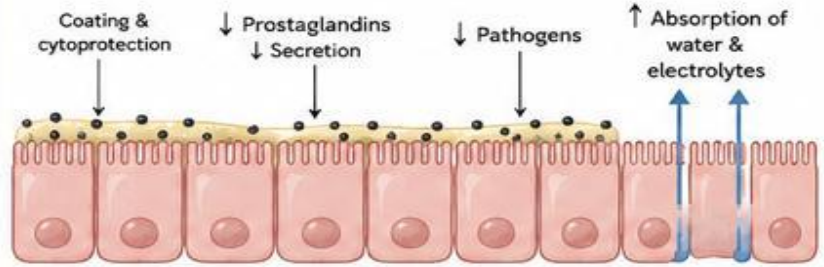
RESULT
Prevents absorption of toxins and microbes
Reduces fluid in stool and intestinal irritation
→ Decreases diarrhea

EXAMPLES
- Activated Charcoal – Adsorbs toxins, gases, microorganisms
- Methyl Cellulose – Adsorbs water, increases stool bulk and consistency

3 FLUID TRANSPORT MODIFIER
(Bismuth Subsalicylate)

MECHANISM
Reduces intestinal secretion and inflammation, promotes absorption of water and electrolytes.

SITE & ACTION
- Coats and protects intestinal mucosa.
- Inhibits prostaglandin synthesis → ↓ intestinal secretion.
- Has mild antimicrobial activity against some enteric pathogens.
- Promotes absorption of water and electrolytes.



RESULT
Decreases secretion and inflammation
Restores fluid and electrolyte balance
→ Decreases diarrhea

EXAMPLE
- **Bismuth Subsalicylate**

OVERALL EFFECT
All three classes act through different mechanisms to reduce stool frequency, improve stool consistency, and restore fluid and electrolyte balance in diarrhea.

Reduced fluid loss | Improved stool consistency | Restored fluid & electrolyte balance

Core Pharmacology References

Lippincott's Pharmacology

Illustrated Reviews: Pharmacology

7th Edition

Rang & Dale's Pharmacology

Rang & Dale's Pharmacology

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